

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses model to form one correct equation (PI by $a = 200$) (ACF) Accept $1.105a + 1.221b = 331$	AO3.1b	M1	$290 = a + b$ $331 = ae^{0.1} + be^{0.2}$
	Forms a second correct equation (ACF)	AO1.1a	M1	$290e^{0.1} = ae^{0.1} + be^{0.1}$ $b = \frac{(331 - 290e^{0.1})}{(e^{0.2} - e^{0.1})} = 90.3$
	Obtains correct a AWRT 200 and b AWRT 90 (AG) Only award if both previous M1's achieved. Do not award marks retrospectively for correct values of a and b used in part (b)	AO1.1b	A1	$= 90$ to the nearest integer so $a = 200$
(b)	Substitutes $t = 3$ and evaluates CAO	AO3.4	B1	$200e^{0.3} + 90e^{0.6} = 434$
(c)	Forms inequality (accept $<$ or $>$) (condone use of equation) FT 'their' value of a , but b must be 90	AO1.1a	M1	$90e^{0.2t} > 200e^{0.1t}$ $e^{0.1t} > \frac{200}{90}$
	Uses logs or calculator to solve 'their' inequality (or equation) If using trial and error must see $t = 7$ and $t = 8$ tested	AO1.1a	M1	$0.1t > \ln\left(\frac{200}{90}\right)$ $t > 10 \ln\left(\frac{200}{90}\right) = 7.985$
	Interprets final result. (Do not accept 2025)	AO3.2a	A1	Just less than 8 so during 2024
(d)	Gives one limitation of the model. Eg. Model must break down as both n_A and n_B will tend to infinity / model assumes nothing changes / no attempt to control the diseases / all the trees have died / finite number of trees / cure for the	AO3.5b	E1	Eventually all of the trees will die so the model will no longer be accurate.

disease might be found / other factors such as drought could affect the model / etc.			
Total 8 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$y = 5 \times 5^x$
Total 1 mark			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses model to form an equation to find k with $t = 5.7$, $m = \frac{1}{2}m_0$	AO3.4	M1	$200 = 400e^{-k \times 5.7}$
	Obtains correct value of k	AO1.1b	A1	
	Uses model to find m with $t = 4$, $m_0 = 400$ and <i>their</i> k (Condone $m_0 = 200$)	AO3.4	M1	$k = 0.1216047...$ $m = 400e^{-0.1216... \times 4}$
	Obtains correct value of m CAO (245.9296...) AWRT 250	AO1.1b	A1	$m = 250$
(b)	Uses model to set up inequality or equation using <i>their</i> k and 280	AO3.1b	M1	$400e^{-0.1216t} \leq 280$ $e^{-0.1216t} \leq 0.7$ $-0.1216t \leq \ln(0.7)$ $t \geq 2.933$
	Solves their inequality or equation to find t (Follow through their k only) (2.933067)	AO1.1b	A1F	
	Interprets <i>their</i> solution (Only follow through if time is earlier than 1:42 pm)	AO3.2a	A1F	10:56 am
(c)	States any sensible reason	AO3.5b	B1	Different people eliminate

such as: Different people eliminate caffeine at different rates The model is based on an average person The length of time taken to drink two cups of coffee may have been significant The amount of caffeine in a “strong cup of coffee” may vary			caffeine at different rates
Total 8 marks			

Q4.

Marking Instructions	AO	Marks	Typical Solution
States the correct gradient of the curve	AO1.2	B1	Grad of curve = $2e^{2x}$
Forms an equation using ‘their’ gradient of the curve and puts it equal to $\frac{1}{2}$	AO1.1a	M1	= grad of tangent so $2e^{2x} = \frac{1}{2}$
Takes a log of each side of ‘their’ equation and uses law of logs to obtain equation in x	AO1.1a	M1	$e^{2x} = \frac{1}{4} \Rightarrow 2x = \ln\left(\frac{1}{4}\right)$
Obtains a correct exact value for x	AO1.1b	A1	$\Rightarrow x = \frac{1}{2} \ln\left(\frac{1}{4}\right) = \ln\left(\frac{1}{2}\right) = -\ln 2$
Substitutes ‘their’ value of x and obtains y value and hence the coordinates (follow through provided values are exact)	AO1.1b	A1F	$y = e^{2x} = \frac{1}{4}$ $\left(-\ln 2, \frac{1}{4}\right)$
Total 5 marks			

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States correct value CAO	AO3.4	B1	50

(i)				
(ii)	States correct integer value CAO	AO3.4	B1	609
(b)	Forms correct equation and rearranges to obtain $e^{0.5t} = \dots$	AO3.4	M1	$150 = 50e^{0.5t}$ so $e^{0.5t} = 3$
	Obtains the correct solution. Must give answer to 3 sf	AO1.1b	A1	$t = 2\ln 3 = 2.20$
(c)	1 mark for any clear valid reason, must be set in context of the question	AO3.5b	E1	No constraint on the number of rabbits (ie could go off to infinity) OR Model is only based on the 3 years of the study. Things may change OR Continuous model but number of rabbits is discrete OR Ignores extraneous factors such as disease, predation, limited food supply
(d)	Forms an equation with exponentials by letting $R = C$ PI	AO3.4	M1	$1000e^{0.1t} = 50e^{0.5t}$ $20 = e^{0.4t}$
	Solves 'their' equation correctly	AO1.1a	M1	$t = \ln 20 \div 0.4$ $= 7.49$
	States correct answer as the year 2023 CAO NMS scores full marks for 2023	AO3.2a	A1	2023
Total 8 marks				

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Makes a deduction about the lower bound of the	AO2.2a	B1	

	function (4)			
	Correctly states the range of f using set notation	AO2.5	B1	The range of f is the set $\{y \in \mathbb{R} : y > 4\}$
(b)	States correctly the set they gave in part (a)	AO1.2	B1F	$\{y \in \mathbb{R} : y > 4\}$
(i)	Interchanges x and y at any stage	AO1.1a	M1	$y = 4 + 3^{-x}$ $x = 4 + 3^{-y}$
(ii)	Rearranges and takes logs	AO1.1a	M1	$3^{-y} = x - 4$ $-y = \log_3(x - 4)$
	Obtains correct expression from completely correct working for $f^{-1}(x)$, notation correct throughout	AO1.1b	A1	$f^{-1}(x) = -\log_3(x - 4)$
(c)	Obtains $gf(x)$	AO1.1b	B1	$gf(x) = g(4 + 3^{-x})$ $= 5 - (4 + 3^{-x})^{0.5}$
(i)	Forms equation and rearranges using 'their' $gf(x) = 2$	AO1.1a	M1	$5 - (4 + 3^{-x})^{0.5} = 2$ $(4 + 3^{-x}) = 9$
(ii)	Correctly rearranges to get a single exponential term where logs can be taken. (Follow through provided 'their' equation requires the use of logs.)	AO1.1b	A1F	$3^{-x} = 5$ $x = -\log_3 5$
	Obtains correct solution	AO1.1b	A1	
Total 10 marks				

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds a difference between 2 terms	AO3.1a	M1	$3e^p - 5 = 5 - 3e^{-p}$ (*) $3e^p - 10 + 3e^{-p} = 0$
	Forms an equation using two differences	AO3.1a	M1	$3e^{2p} - 10e^p + 3 = 0$
	Forms a quadratic equation in e^p	AO1.1a	M1	$e^p = \frac{1}{3}, 3$
	Obtains a correct quadratic equation	AO1.1b	A1	$p = \ln \frac{1}{3}, \ln 3$ ALT to (*)

(b)	Obtains 2 correct solutions for e^p from 'their' quadratic FT only applies if previous mark has been awarded	AO1.1b	A1F	$2(5 - 3e^{-p}) = 3e^p - 3e^{-p}$ Or $2(3e^{-p} - 5) = 3e^p - 3e^{-p}$
	Obtains final answers in an exact form FT applies if previous mark has been awarded	AO2.2a	A1F	
	Finds a ratio between two consecutive terms (no requirement to use a and r)	AO3.1a	M1	Assume it is possible that $3e^{-q}$, 5 and $3e^q$ are three consecutive terms of a geometric sequence $a = 3e^{-q}$, $ar = 5$, $ar^2 = 3e^q$
	Compares two ratios (could be ratios of successive terms, no requirement to use a and r)	AO3.1a	M1	$\frac{ar}{a} = \frac{5}{3e^{-q}} \Rightarrow r = \frac{5e^q}{3}$ $\frac{ar^2}{ar} = \frac{3e^q}{5} \Rightarrow r = \frac{3e^q}{5}$
	Identifies a contradiction	AO2.1	R1	$\frac{5}{3e^{-q}} = \frac{3e^q}{5} \Rightarrow 25 = 9$
	Draws a conclusion about k	AO2.4	R1	This is a contradiction therefore $3e^{-q}$, 5 and $3e^q$ cannot form three consecutive terms of a geometric sequence.
				Total 10 marks

Q8.

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds 2 nd derivative and sets up an inequality	AO3.1a	M1	$\frac{dy}{dx} = -2xe^{-x^2}$
	Obtains correct first derivative	AO1.1b	A1	$\frac{d^2y}{dx^2} = -2e^{-x^2} + 4x^2e^{-x^2}$
	Obtains second derivative correct from 'their' first derivative	AO1.1b	A1F	$-2e^{-x^2} + 4x^2e^{-x^2} < 0$ $4x^2 - 2 < 0$
	Deduces correct final inequality (could use set notation)	AO2.2a	A1	$-\frac{\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}$

(b)	Uses trapezium rule	AO1.1a	M1	$\int_{0.1}^{0.5} e^{-x^2} dx \approx \frac{0.1}{2} (e^{-0.01} + e^{-0.25})$ $+ 2(e^{-0.04} + e^{-0.09} + e^{-0.16})$ ≈ 0.3611
	Trapezium rule entries all correct	AO1.1b	A1	
	Finds correct value	AO1.1b	A1	
(c)	References area being completely within concave section So...	AO2.4	E1	$[0.1, 0.5] \subset \left[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right]$ $\therefore$ area is completely within concave section Hence trapezia lie below curve and give an under-estimate for the area
	Trapezia all fall completely underneath the curve therefore underestimate (only award this mark if previous E1 has been awarded)	AO2.4	E1	
(d)	Uses suitable rectangle to obtain over-estimate	AO3.1a	B1	Using a rectangle with the left hand edge the same height as the curve will produce an over-estimate Area of rectangle = $0.4 \times e^{-0.01^2} = 0.396...$ $\therefore 0.36 < A < 0.40$ So $A = 0.4$ to 1 dp
	Explains that this rectangle lies above the curve	AO2.4	E1	
	Constructs rigorous mathematical argument about accuracy, which leads to required result Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
				Total 12 marks

©2025 Exam Papers Practice. All rights reserved.

Q9.

- (a) Stretch(I) in y-direction(II) scale factor 3(III)

OE Need (I) and either (II) or (III)

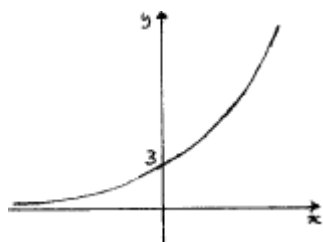
M1

All correct. Need (I) and (II) and (III)
[>1 transformation scores 0 / 2]

A1

2

- (b)



Shape with indication of correct asymptotic behaviour in 2nd quadrant below pt of intersection with y-axis

B1

Only intersection is with y-axis, and only intercept is 3 stated / indicated

B1

2

(c) $3 \times 4^x = 4^{-x}$

OE eqn. in x

M1

$$\log 3 + \log 4^x = \log 4^{-x}$$

Log Law 1 (or Law 2 applied to $\frac{4^x}{4^{-x}} = 3$ or $\frac{1}{3}$ OE) used correctly or correct rearrangement to $4^{2x} = 1/3$ OE simplified e.g. $16^x = 3^{-1}$ or $4^x = (1/\sqrt{3})$

m1

$$\log 3 + x \log 4 = -x \log 4$$

Log Law 3 applied correctly twice (dependent on both M1 & m1) or a correct method using logs to solve an eqn. of form $a^{kx} = b$, $b > 0$ (including case $k = 1$) (dependent on M1 and valid method to a^{kx})

m1

$$x = \frac{-\log 3}{2 \log 4} \quad \left(= \frac{-\log 3}{\log 16} \right)$$

Correct expression for x or for -x

e.g. $x = \frac{1}{2} \log_4 \left(\frac{1}{3} \right)$

PI by correct 3sf value or better

A1

$$x = -0.396(2406 \dots) = -0.396 \text{ (to 3sf)}$$

If logs not used explicitly then max of M1m1m0.

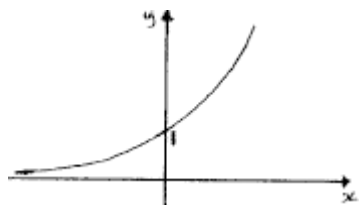
A1

5

[9]

Q10.

(a)



*Correct graph, must clearly go below the intersection pt and an indication of correct behaviour of curve for large positive and large negative values of x .
Ignore any scaling on axes.*

B1

Only one y -intercept, marked / stated as 1 or as coords $(0, 1)$ with graph having no other intercepts on either axes.

B1

2

(b) $9^x = 15 \Rightarrow x \log 9 = \log 15$

OE eg $x = \log_9 15$

M1

$(x =) 1.23(2486...) = 1.23$ to 3sf

Condone > 3sf.

Must see evidence of logs used so NMS scores 0/2

A1

2

(c) $\{f(x) =\} 9^{-x}$

OE

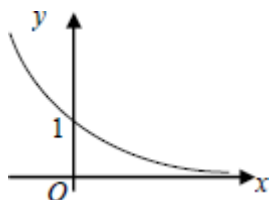
B1

1

[5]

Q11.

(a)



Correct shaped graph in 1st two quadrants only and indication of correct behaviour of curve for large positive and negative vals. of x . Ignore any scaling on axes.

B1

y -intercept indicated as 1 on diagram or stated as intercept = 1 or as coords (0, 1).

B1

2

(b) $\frac{1}{2^x} = \frac{5}{4} \Rightarrow 2^{-x} = \frac{5}{4}$ (or $2^x = \frac{4}{5}$ or $2^{2-x} = 5$)

Correct 'rearrangement' to eg $2^x = \frac{4}{5}$ or $2^{-x} = \frac{5}{4}$ or $0.5^x = 1.25$ PI or $\log 1 - \log 2^x = \log(5/4)$ or better

M1

$$\log 2^{-x} = \log 1.25 \Rightarrow -x \log 2 = \log 1.25$$

$$[\log 2^x = \log 0.8 \Rightarrow x \log 2 = \log 0.8]$$

$$[\log 2^{2-x} = \log 5 \Rightarrow (2-x) \log 2 = \log 5]$$

$$[2^x = 0.8, x \log_2 0.8]; [0.5^x = 1.25, x = \log_{0.5} 1.25]$$

Takes logs of both sides of eqn of form either $2^x = k$ or $2^{-x} = k$ OE and uses 3rd law of logs or log to base 2 (or base $\frac{1}{2}$) correctly

M1

$$x = -0.321928 \dots \text{ so } x = -0.322 \text{ (to 3sf)}$$

Condone > 3sf [Logs must be seen to be used otherwise **max** of M1M0A0]

A1

3

(c) $\log_a b^2 + 3 \log_a y = 3 + 2 \log_a \left(\frac{y}{a} \right)$

$$\log_a b^2 + 3 \log_a y = 3 + 2 [\log_a y - \log_a a]$$

A log law used correctly; condone missing base a .

M1

$$\log_a b^2 + 3\log_a y = 3 - 2\log_a a$$

A different log law used correctly condone missing base a.

M1

$$\log_a b^2 y = 3 - 2(1) \quad [\text{or } \log_a b^2 y + \log_a a^2 = 3]$$

Either a further different log law used correctly condone missing base a or $\log_a a = 1$ a stated / used.

M1

$$\Rightarrow \log_a b^2 y = 1 \quad \Rightarrow b^2 y = a$$

$\log Z = k \Rightarrow Z = a^k$ used or a correct method to eliminate logs (dep on no misapplication of any log law OE in the whole solution) Rearrangements which require only two of the above Ms to eliminate logs correctly: award the remaining M with the m mark.

M1

$$\Rightarrow y = ab^{-2}$$

ACF of RHS

A1

5

[10]

Q12.

(a) $P = 500e^{\frac{1}{8} \times 60}$

Must use $t = 60$

M1

$$= 904\,000$$

Nearest thousand required 904000 only

A1

2

(b) (i) $\left(e^{\frac{1}{8}t}\right)^2 = \frac{500000}{500}$

M1

$$t = 8 \ln \sqrt{1000}$$

OE Take logs correctly leading to expression for t.

M1

$t = 27.6$ (minutes)
 Accept 27.631

A1

3

Alternative

$$e^{\frac{1}{8}t} = 1000e^{-\frac{1}{8}t} \Rightarrow e^{\frac{1}{4}t} = \frac{500000}{500}$$

(M1)

$$t = 4 \ln 1000$$

Take logs correctly leading to expression for t .

(M1)

$$t = 27.6 \text{ (minutes)}$$

(A1)

(3)

Alternative

$$e^{\frac{1}{8}t} = 1000e^{-\frac{1}{8}t} \Rightarrow \ln\left(e^{\frac{1}{8}t}\right) = \ln 1000 + \ln\left(e^{-\frac{1}{8}t}\right)$$

Take logs correctly.

(M1)

$$t = 4 \ln 1000$$

(M1)

$$t = 27.6 \text{ (minutes)}$$

(A1)

27.6 as final answer NMS 3/3

27.6 following wrong working AO (FIW) but could still score M mark(s)

(3)

(ii) $500e^{\frac{1}{8}t} - 500000e^{-\frac{1}{8}t} = 45000$

Set up equation; condone one error; allow in t .
 Condone inequality.

M1

$$\times \frac{e^{\frac{1}{8}t}}{500} \Rightarrow \left(e^{\frac{1}{8}t}\right)^2 - 1000 = 90e^{\frac{1}{8}t}$$

$$\left(e^{\frac{1}{8}t}\right)^2 - 90e^{\frac{1}{8}t} - 1000 = 0$$

Multiply by $\frac{e^{\frac{1}{8}t}}{500}$ and rearrange to AG, be convinced.

A1

$$e^{\frac{1}{8}t} = 100 \quad (e^{\frac{1}{8}t} = -10 \text{ rejected})$$

Solve quadratic equation (retaining positive root).

M1

$$t = 36.8 \text{ (minutes)}$$

CAO

A1

M1 for solve quadratic equation. Let $x = e^{\frac{1}{8}t}$ solve quadratic equation $x^2 - 90x - 1000 = 0$ by inspection, $x = 100$ seen; factors $(x - 100)(x + 10)$ with 100 and 10 seen; complete square $x = 45 \pm \sqrt{3025}$ all correct.

$$\text{formula} \quad \frac{90 \pm \sqrt{90^2 + 4000}}{2} \quad \text{all correct.}$$

Final answer must have $t = 36.8$ for A1

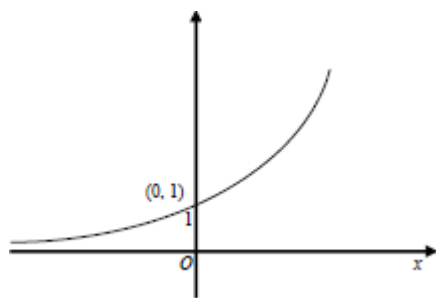
©2025 Exam Papers Practice. All rights reserved.

4

[9]

Q13.

(a)



Correct shape, curve in 1st two quadrants only, crossing positive y-axis once and asymptotic to negative x-axis.

B1

Coordinates (0, 1). Accept y-intercept indicated as 1 on diagram or stated as 'intercept = 1' B0 if graph clearly drawn crossing axes at more than one point

B1

2

(b) (i) $y^2 - 12 = y$ OE; $7^{2x} - 12 = 7^x$ OE

Eliminates either x or y correctly

M1

$(y - 4)(y + 3) (= 0); (7^x - 4)(7^x + 3) (= 0)$

Correct factors or $y = \frac{1 \pm \sqrt{49}}{2}$ or better or $7^x = \frac{1 \pm \sqrt{49}}{2}$ or better

A1

Since $y (= 7^x) > 0$, $[y (= 7^x) \neq -3]$
(there is exactly one point of intersection)

Clear indication that c's negative solution(s) has / have been considered and rejected

E1

y-coordinate is 4

B1

4

(ii) $7^x = 4$ so $x \log 7 = \log 4$ [or $x = \log_7 4$]

OE ft on $7^x = k$, where k is positive, to either $x \log 7 = \log k$ or $x = \log_7 k$

M1

$x = 0.712(414...) = 0.712$ to 3SF

Condone > three significant figures. If use of logarithms not explicitly seen then score 0 / 2

A1

2

[8]

Q14.

(a) (i) $(p =) 3$

B1

1

(ii) $(q =) -3$

If not correct, ft on $-p$

B1F

1

(iii) $(r =) \frac{1}{2}$

OE

B1

1

(b) $2^{\frac{1}{2}} \times 2^x = 2^{-3} \Rightarrow 2^{\frac{1}{2}+x} = 2^{-3}$

Using a law of indices or logs correctly to combine at least two of the powers of 2 PI

M1

$\Rightarrow x = -3\frac{1}{2}$

If not correct, ft on $x = q - r$ provided method shown

A1F

2

[5]

Q15.

(a) $m = 10 \times 2^{-\frac{14}{8}}$

©2025 Exam Papers Practice. All rights reserved.

M1

$\approx 3 \text{ (gm)}$

Condone 2.97 or better

NOT 2.9 as final answer

A1

2

(b) $2^{-\frac{d}{8}} = \frac{1}{16}$

M1

$\frac{d}{8} = 4 \Rightarrow d = 32$

CSO

A1

2

(c) $0.01m_0 = m_0 \times 2^{-\frac{t}{8}}$

m_0 can be numerical

M1

$$\ln(0.01) = -\frac{t}{8} \ln(2)$$

Take logs correctly from their equation leading to a linear equation in t .

M1

$$t = 53.15$$

$$n = 54$$

CSO

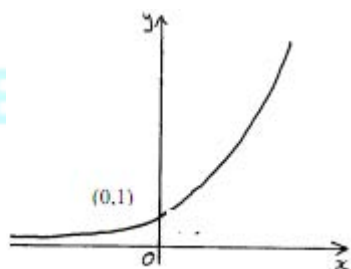
A1

3

[7]

Q16.

(a)



Any graph only crossing the y -axis at $(0, 1)$ stated /indicated ... (accept 1 on y -axis as equivalent) and not drawn **below** x -axis

B1

Correct shaped graph, must clearly go below the intersection pt and an indication of correct behaviour of curve for large positive and large negative values of x . Ignore any scaling on axes.

B1

2

(b) Translation;

*Accept 'transl...' as equivalent
[T or Tr is NOT sufficient]*

B1

$$\begin{bmatrix} 0 \\ -5 \end{bmatrix}$$

*If vector not given, accept **full** equivalent to vector in words provided linked to 'transl../move/shift'
(B0B0 if >1 transformation)*

B1

2

(c) (i) $4^x = (2^2)^x = 2^{2x} = (2^x)^2 = Y^2$

$$2^{x+2} = 2^x \times 2^2 = 4Y$$

Justifying either $4^x = Y^2$ or $2^{x+2} = 4Y$ with no errors seen

M1

$$4^x - 2^{x+2} - 5 = 0 \Rightarrow Y^2 - 4Y - 5 = 0$$

AG Be convinced; must have justified both of the above.

A1

2

(ii) $(Y - 5)(Y + 1) = 0$

Correct factorising or use of quadratic formula or completing sq. Pl by both solns 5 & -1 seen

M1

(Since) $2^x > 0$ (for all real x), $2^x = 5$ so only one (real) solution

Rejection of 2^x (condone Y) negative, with justification, (condone " 2^x not negative") followed by statement

E1

$$\log 2^x = \log 5 \Rightarrow x \log 2 = \log 5$$

Eqn of form $p^x = q \Rightarrow x \log p = \log q$ provided $p > 0$ & $q > 0$ OE eg $x = \log_2 5$

M1

$$x = 2.3219... = 2.322 \text{ (to 3dp)}$$

Condone > 3dp but must see explicit use of logs and must only be the one solution.

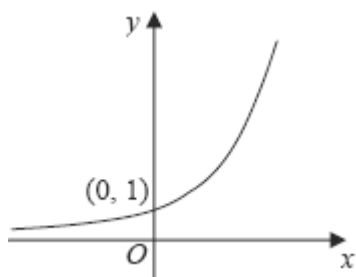
A1

4

[10]

Q17.

(a)



Shape with some indication of asymptotic behaviour in 2nd quadrant below pt of intersection with y-axis

B1

Only intersection is with y-axis at (0, 1) stated/indicated ... (accept 1 on y-axis as equivalent)

B1

2

(b) (i) $h = 0.5$
PI

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

B1

$$f(x) = 2^x$$

$$I \approx h/2 \{ \dots \}$$

$$\{ \dots \} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$$

OE summing of areas of the 4 'trapezia'

M1

$$\{ \dots \} = 1 + 4 + 2(\sqrt{2} + 2 + \sqrt{8})$$

$$= 5 + 2 \times 6.2426... = 17.485...$$

OE Accept 2 dp (rounded or truncated) as evidence for surds

A1

$$(I \approx) 4.3713... = 4.37 \text{ (to 3 sf)}$$

CAO Must be 4.37
 SC for those who use 5 strips, max
 possible is B0M1A1A0

A1 4

- (ii) Increase the number of ordinates
OE

E1 1

- (c) Translation;
Accept 'translat...' as equivalent
[T or Tr is NOT sufficient]

B1;

$$\begin{bmatrix} -7 \\ 3 \end{bmatrix}$$

B1 for each component of the vector.
 Condone if the equiv 2 vectors are given.
 Accept **full** equivalent to vector(s) in
 words provided linked to 'translation/
 move/shift' and **correct** directions.
 (No marks if **different** transformations)

B1;B1 3

- (d) $8 = 2^k + 3 \Rightarrow 2^k = 5$

Correct subst. and an attempted

rearrangement to $2^k = N$. PI by $k = \frac{\log 5}{\log 2}$

M1

$$k = \log_2 5$$

Accept $m = 2, n = 5$

A1 2

[12]

Q18.

- (a) $(y =) 1$

B1

(b) $h = 0.2$

PI

B1

$$f(x) = 2^{4x}$$

$$I \approx h/2 \{...\}$$

$$\{.\} = f(0) + f(1) + 2[f(0.2) + f(0.4) + f(0.6) + f(0.8)]$$

OE summing of areas of the 'trapezia'..

M1

$$\{.\} = 1 + 16 + 2(2^{0.8} + 2^{1.6} + 2^{2.4} + 2^{3.2})$$

$$= 1 + 16 + 2(1.741.. + 3.031.. + 5.278.. + 9.1895..)$$

$$= [17 + 2 \times 19.24...]$$

OE Accept 2dp rounded or truncated evidence

A1

$$I = 5.55 \text{ (to 2 dp)}$$

Must be 5.55

A1

4

(c) Stretch(I) in y -direction(II) scale factor $\frac{1}{8}$ (III)

*Need (I) **and** either (II) or (III)*

M1

Need (I) and (II) and (III)

A1

ALTn: Translation with an indication
that the translation is in the x -direction (B1)

$$\begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$$

(B1)

*Combination of **different**
transformations scores 0/2*

2

(d) $g(x) = 2^{4(x-1)} - \frac{1}{2}$

B1 for either $2^{4(x+1)} - \frac{1}{2}$ or for

$$2^{4(x-1)} + \frac{1}{2} \text{ or for } 2^{4x-1} - \frac{1}{2}$$

B2,1,0

At Q , $y = 0 \Rightarrow 2^{4(x-1)} = 2^{-1}$

Reaches a stage from which linear eqn can be stated directly eg an alternative stage is

$$4(x-1) \log 2 = -\log 2$$

M1

$$\Rightarrow 4x - 4 = -1 \Rightarrow x = 0.75$$

NMS mark as 4 or 0

A1

4

(e) (i) $\log_a k = \log_a 2^3 + \log_a 5 - \log_a 4$

$$\log_a k = \log_a (2^3 \times 5) - \log_a 4$$

One law of logs used

M1

A second law of logs used; could be

$$\log_a k = \log_a 2^3 + \log_a \left(\frac{5}{4}\right)$$

M1

$$\log_a k = \log_a \left(\frac{2^3 \times 5}{4}\right) = \log_a 10 \Rightarrow k = 10$$

CSO AG

A1

3

(ii) $2^{4x-3} = \frac{5}{4}$ so

*Equate y's, take logs (to any base) of both sides **and** apply 3rd law of logs.*

$$(4x-3) \log_{10} 2 = \log_{10} \frac{5}{4}$$

M1

$$x = \frac{3 \log_{10} 2 + \log_{10} \left(\frac{5}{4} \right)}{4 \log_{10} 2}$$

m1

$$x = \frac{\log_{10} 10}{4 \log_{10} 2} \text{ so } x = \frac{1}{4 \log_{10} 2}$$

A1

Altn $4x \log 2 = \log \left(\frac{5}{4} \times 2^3 \right)$

Rearrange correctly to $x = \dots$

Altn $4x \log 2 = \log 10$

In both cases, log term(s) must have same base and expressions must be in an exact form, ie not approx. dec. vals

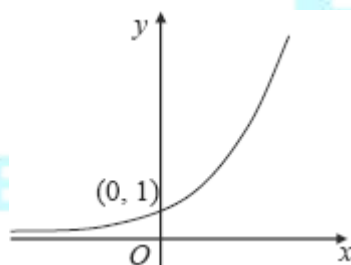
CSO AG Must be clear evidence that base 10 is used, also be convinced

3

[17]

Q19.

(a)



Shape (graph must clearly go below the intersection pt.). Condone if x -axis is a tangent

B1

Only intersection with y -axis at $(0, 1)$ stated/indicated ... (accept 1 on y -axis as equivalent) 0

B1

2

- (b) (i) Stretch **(I)** in x -direction **(II)** scale factor 0.5 **(III)**
Need **(I)** & one of **(II)**, **(III)**

M1

M0 if >1 transformation

A1

2

(ii) Translation;

Must be 'Translation' or 'translate (d)' for 1st B mark

B1;

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

*Accept **full** equivalent to vector in words provided linked to 'translation/ move/shift' and **negative** x-direction (Note: B0 B1 is possible)*

B1

ALTn: Stretch (I) in y-direction (II) scale factor 3 (III)

[Mark the alternative as in (b)(i).]

2

(c) (i) $9^x = (3^2)^x = 3^{2x} = (3^x)^2 = Y^2;$

Justifying either $9^x = Y^2$ or $3^{x+1} = 3Y$

M1

$$3^{x+1} = 3^x \times 3^1 = 3Y$$

$$9^x - 3^{x+1} + 2 = 0 \Rightarrow Y^2 - 3Y + 2 = 0$$

$$\Rightarrow (Y-1)(Y-2) = 0$$

AG

EXAM PAPERS PRACTICE

A1

2

(ii) $Y = 1 \Rightarrow 3^x = 1 \Rightarrow x = 0$

AG (Accept direct substitution if convinced)

B1

$$Y = 2 \Rightarrow 3^x = 2$$

$$\log_{10} 3^x = \log_{10} 2$$

Takes logs of both, PI by 'correct' value(s) later.

or $x = \log_3 2$ seen

M1

$$x \log_{10} 3 = \log_{10} 2$$

Use of $\log 3^x = x \log 3$ or

$$\log_3 2 = \frac{\lg 2}{\lg 3} \quad \text{OE (PI by } \log_3 2 = 0.630 \text{ or } 0.631 \text{ or better)}$$

m1

$$x = \frac{\lg 2}{\lg 3} = 0.630929 \dots = 0.6309 \text{ to 4 dp}$$

Must show that logarithms have been used otherwise 0/3

A1

4

[12]

Q20.

- (a) (i) $h = 0.5$
 Integral = $\frac{h}{2} \{ \dots \}$
 PI

B1

$$\{ \dots \} = f(0) + 2\left[f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{3}{2}\right)\right] + f(2)$$

OE summing of areas of the four traps.

M1

$$\{ \dots \} = 1 + 2[\sqrt{6} + 6 + 6\sqrt{6}] + 36$$

$$= 1 + 2[2.449.. + 6 + 14.6969..] + 36$$

Condone 1 numerical slip. Accept 3sf values if not exact.

A1

$$= 37 + 2 \times 23.146.. = 83.292..$$

$$\text{Integral} = 0.25 \times 83.292.. = 20.8 \text{ (3sf)}$$

CAO; must be 20.8

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph
Accept single trapezium with its sloping side above the curve

M1

{Approximation is an} overestimate

Dep. on 4 trapezia with each of their upper vertices lying on the curve

A1
2

- (b) (i) Stretch (I) in x -direction (II)
Need (I) and one of (II), (III)
M0 if more than one transformation

M1

(scale factor) $\frac{1}{3}$ (III)

A1
2

- (ii) $6^{3x} = 84$
PI

M1

$\log_{10} 6^{3x} = \log_{10} 84$
Take logs of both sides of $a^x = b$, PI by 'correct' value(s) later or $3x = \log_6 84$

M1

$3x \log_{10} 6 = \log_{10} 84$
Use of $3 \log 6^{3x} = 3x \log 6$ OE
or $3x = \log_6 84$ seen

m1

$$x = \frac{\lg 84}{3 \lg 6}$$

$x = 0.82429\dots = 0.824$ (to 3dp)
Must see that logs have been used before any of the last 3 marks are awarded in (b)(ii). Condone > 3dp

A1
4

- (c) $f(x) = 6^{x-1} - 2$
B1 For either $6^{x-1} + 2$ or for $6^{x+1} - 2$

B2,1
2

Q21.

(a) $y_A = 3(2^0 + 1)$

Substituting $x = 0$ PI

M1

$= 6$

A1

2

(b) $h = 2$

PI

B1

$\text{Integral} = h/2 \{ \dots \}$

$\{ \dots \} = f(0) + 2[f(2) + f(4)] + f(6)$

OE summing of areas of the three traps.

M1

$\{ \} = 6 + 2[3 \times 5 + 3 \times 17] + 3 \times 65 = 6 + 2[15 + 51] + 195$

Condone 1 numerical slip {ft on (a) for $f(0)$ if not recovered}

[Sum of 3 traps. = 21 + 66 + 246]

A1

$\text{Integral} = 333$

CAO

A1

4

(c) (i) $21 = 3(2^x + 1) \Rightarrow 2^x = 6$

AG (be convinced)

B1

1

(ii) $\log_{10} 2^x = \log_{10} 6$

*Take $\ln$ or $\log_{10}$ of both sides of $a^x = b$
or other relevant base if clear. The
equation $a^x = b$ used must be correct.*

M1

$x \log_{10} 2 = \log_{10} 6$

Use of $\log 2^x = x \log 2$ OE

m1

$$x = \frac{\lg 6}{\lg 2} = 2.5849\dots = 2.58 \text{ to 3sf}$$

Both method marks must have been awarded.

A1

3

[10]

Q22.

(a) (i) $t = 0: x = 3$

B1

1

(ii) $t = 14: x = 15 - 12e^{-1}$

or $15 - 12e^{\frac{-14}{14}}$

$= 10.6$

CAO

M1

A1

2

(b) (i) $-5 = -12e^{-\frac{t}{14}}$

substitute $x = 10$; rearrange to form $p = qe^{-\frac{t}{14}}$

M1

$$\ln\left(\frac{5}{12}\right) = -\frac{t}{14} \quad (\text{OE})$$

take lns correctly

m1

$$t = 14 \ln\left(\frac{12}{5}\right)$$

must come from correct working

A1

3

(ii) $t = 12.256... \approx 12$ days

*ft on a, b if a > b; accept t = 12 NMS
 Accept 12 from incorrect working in (b)(i)
 Accept 13 if 12.2 or 12.3 seen*

B1F

1

(c) (i) $\frac{dx}{dt} = -\frac{1}{14}x - 12e^{-\frac{t}{14}}$

differentiate; allow sign error

condone $\frac{dy}{dx}$ used consistently

M1

$$= -\frac{1}{14}(x - 15)$$

Or $\frac{1}{14}\left(12e^{-\frac{t}{14}}\right)$ and $12e^{-\frac{t}{14}} = 15 - x$ seen

m1

$$= \frac{1}{14}(15 - x)$$

AG – be convinced CSO

A1

3

Alt: $t = -14 \ln\left(\frac{15 - x}{12}\right)$

attempt to solve given equation for t

(M1)

$$\frac{dt}{dx} = \frac{-14\left(-\frac{1}{12}\right)}{\left(\frac{15 - x}{12}\right)}$$

differentiate wrt x, with $\frac{1}{15 - x}$ seen; OE

(m1)

$$\frac{dt}{dx} = \frac{14}{15-x} \Rightarrow \frac{dx}{dt} = \frac{1}{14}(15-x)$$

AG – be convinced

(A1)

(3)

Alt: (backwards)

$$\int \frac{dx}{15-x} = \int \frac{dt}{14} = \pm 14 \ln(15-x) = t + c$$

(M1)

Use (0,3): $-14 \ln(15-x) + 14 \ln 12 = t$

(m1)

Solve for x : $x = 15 - 12e^{-\frac{t}{14}}$

All steps shown

(A1)

(3)

(ii) rate of growth = 0.5 (cm per day)

Accept $\frac{7}{14}$

B1

1

[11]

Q23.

(a) $\log 0.8^x = \log 0.05$ OR $x = \log_{0.8} 0.05$ **(M1)**

NMS: SC B2 for 13.425 or better
(B1 for 13.4 or 13.43; 13.42)

M1

$x \log_{10} 0.8 = \log_{10} 0.05$ oe

A1

$x = 13.425$ to 3dp OR 13.425 **(A2)**
(else A1 for 1 or 2dp)

Condone greater accuracy

A1

3

(b) (i) $\frac{a}{1-r}$

$$S_{\infty} = \frac{a}{1-r} \text{ used}$$

M1

$$\frac{a}{1-r} = 5a \Rightarrow a = 5a(1-r)$$

Or better

A1

$$\Rightarrow 1 = 5(1-r) \Rightarrow r = \frac{4}{5} = 0.8$$

AG (be convinced)

A1

3

(ii) $n^{\text{th}} \text{ term} = 20 \times (0.8)^{n-1}$

Condone $20 \times (0.8)^{n-1}$

M1

$$n^{\text{th}} \text{ term} < 1 \Rightarrow 0.8^{n-1} < \frac{1}{20} \text{ oe}$$

$$0.8^{n-1} < 0.05 \text{ or } 0.8^{n-1} = k, \text{ where } k = 0.05$$

or k rounds **up** to 0.050

A1

Least n is 15

If not 15, ft on integer part of [answer (a)+2]
provided $n > 2$

A1F

SC 3/3 for 15 if no error

SC $n^{\text{th}} \text{ term} = 16^{n-1}$ M1A0A0

3

[9]

Q24.

(a) $a = -8$

B1

$$e^{2x} - 9 = 0$$

M1

$$e^{2x} = 9$$

$$2x = \ln 9$$

$$x = \ln 3$$

AG Condone verification

A1

3

(b) $(e^{2x} - 9)^2 = e^{4x} - 18e^{2x} + 81$

AG

B1

1

[4]

Q25.

(a) $y = x^{-2} \ln x$

$$\frac{dy}{dx} = x^{-2} \frac{1}{x} - 2x^{-3} \ln x$$

Use of product or quotient

M1

each term

A1 A1

$$= \frac{1 - 2 \ln x}{x^3}$$

Convincing argument $x^{-2} \times \frac{1}{x} = x^{-3}$

AG

A1

4

(b) $\int x^{-2} \ln x \, dx$ $u = \ln x$ $dv = x^{-2}$

Attempt at integration by parts

M1

$$du = \frac{1}{x} \quad v = -x^{-1}$$

A1

$$\int = -\frac{1}{x} \ln x + \int x^{-2} dx$$

A1

$$= -\frac{1}{x} \ln x - \frac{1}{x} (+C)$$

A1

4

(c) (i) At A, $\frac{dy}{dx} = 0$

$$1 - 2 \ln x = 0$$

$$\ln x = \frac{1}{2}$$

Attempt at $\ln x = k$

M1

$$x = e^{\frac{1}{2}}$$

A1

2

(ii) $R = \left[-\frac{1}{x} (\ln x + 1) \right]_1^5$
 $R = [\text{Their (b)}]_1^5$

M1

$$= -\frac{1}{5} (\ln 5 + 1) + (\ln 1 + 1)$$

OE

A1

$$= \frac{1}{5} (4 - \ln 5)$$

convincing argument; AG

A1

3

[13]

Q26.

(a) $A = 80$

B1

1

(b) $5000 = 80 \times k^{56}$

SC1 Verification. Need 62.51 or better

M1

$$k = \sqrt[56]{\frac{5000}{80}} \approx 1.07664$$

Or using logs: M1 $\ln\left(\frac{5000}{80}\right) = 56 \ln k$

$$A1 \ k = e^{\frac{\ln\left(\frac{5000}{80}\right)}{56}}$$

Or 3/3 for $k = 1.076636$

Or 1.076637 seen

M1A1

3

(c) (i) $V = 80 \times k^{106} = 200707$

200648 using full register k

M1A1

2

(ii) $\ln 10000 = \ln k^t$

M1

$$t = \frac{\ln 10000}{\ln k} = 124.7 \Rightarrow 2024$$

M1 $t \ln k = \ln 10000$

A1 CAO

Or trial and improvement M1 expression

M1 125, 124, A1 2024

M1A1

3

[9]

Q27.

(a) (i) y -coordinate of A is $27 - 3^0 = 26$

M1A1

2

- (ii) When $x = 3, y = 27 - 3^3 = 0 \Rightarrow B(3,0)$
AG; be convinced

B1

1

- (b) $h = 1$

PI

B1

Area $\approx h/2\{\dots\}$
 $\{\dots\} = f(0) + f(3) + 2[f(1) + f(2)]$
OE summing of areas of the 'trapezia'..

M1

$\{\dots\} = "26" + 0 + 2(24 + 18)$
on (a)(i) ($\Sigma trap = "25" + 21 + 9$)

A1 ✓

(Area $\approx$) 55
on $[42 + 0.5 \times "(a)(i)"]$

A1 ✓

4

- (c) (i) $\log_{10} 3^x = \log_{10} 13$
Takes $\ln$ or $\log_{10}$ on both or $x = \log_3 13$

M1

$$x \log_{10} 3 = \log_{10} 13$$

Use of $\log 3^x = x \log 3$ or $\log_3 13 = \frac{\lg 13}{\lg 3}$ OE
(PI by $\log_3 13 = 2.335$ or better)

m1

$$x = \frac{\lg 13}{\lg 3} = 2.334717$$

.... = 2.3347 to 4dp

Must show that logarithms have been used

A1

3

- (ii) $\{k \Rightarrow\} 14$

Condone $y = 14$; Accept final answer 14 with only zeros after decimal point e.g. 14.000

B1

1

(d) (i) Translation;

'Translation' / 'translate(d)'

B0 if more than one transformation

B1

$$\begin{bmatrix} 0 \\ -27 \end{bmatrix}$$

Accept full equivalent in words provided

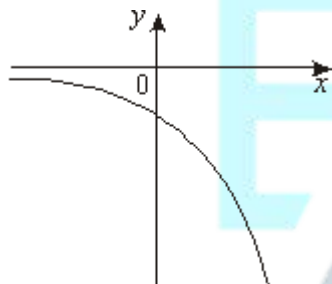
linked to 'translation/move/shift' and negative y -direction

(Note: B0 B1 is possible)

B1

2

(ii)



Correct shape (translation of given curve vertically downwards)

B1

Only point of intersection with coordinate axes is on negative y -axis and curve is asymptotic to the negative x -axis

B1

2

[15]

Q28.

(a) $f(x) = 2e^{3x} - 1$

Range: $f(x) > -1$ (or $y > -1$ or $f > -1$)
for -1 only

M1

exactly correct

A1

2

(b) $y = 2e^{3x} - 1$
 $x = 2e^{3y} - 1$
 $x \leftrightarrow y$

M1

$$2e^{3y} = x + 1$$

$$e^{3y} = \frac{x+1}{2}$$

attempt to isolate

M1

$$y = \frac{1}{3} \ln \left(\frac{x+1}{2} \right)$$

all correct with no error AG (be convinced)

A1

3

(c) $f^{-1}(x) = \frac{1}{3} \left(\frac{2}{x+1} \right) \times \frac{1}{2}$ OE

for differentiation of $\ln$; $\frac{k}{\ln(x \pm 1)}$

M1

for $\frac{1}{2}$

A1

all correct

A1

$$x = 0$$

$$f^{-1}(x) = \frac{1}{3}$$

CSO

A1

4

Alternative

$$f^{-1}(x) = \frac{1}{3} \ln(x+1) - \frac{1}{3} \ln 2$$

M1A1

$$f^{-1}(x) = \frac{1}{3(x+1)}$$

A1

$$f^{-1}(0) = \frac{1}{3}$$

CSO

A1

[9]

Q29.

- (a) {y-coordinate of A is} 2

B1

1

- (b) (i) h = 0.25

B1

$$\text{Integral} = \frac{h}{2} \{...\}$$

$$\{...\} =$$

$$f(0) + 2\left[f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)\right] + f(1)$$

$$\{...\} =$$

M1

$$2 + 4 + 2[(2.316.. + 2.732.. + 3.279(5..)]$$

A1 ✓

$$\{= 6 + 2 \times 8.3276..\} \{= 22.65(5...)\}$$

Condone one numerical slip
Accept values to 3sf (rnd or trunc)
ft answer from (a) if not "2"

$$\text{Integral} = 0.125 \times 22.655.. = 2.8319..$$

$$\text{Integral} = 2.83 \text{ to 3 sf}$$

CAO Must be 2.83

(NMS scores 0/4)

A1

4

(ii) Relevant trapezia drawn on a copy of given graph

Accept relevant single trapezium with its sloping side above the curve

M1

{Approximation is an}overestimate

A1

2

(c) $5 = 3^x + 1 \Rightarrow 3^x = 4$

B1

$$\log_{10} 3^x = \log_{10} 4$$

Takes ln or $\log_{10}$ on both sides of $3^x = k$, where $k > 0$

M1

$$x \log_{10} 3 = \log_{10} 4$$

Use of $\log 3^x = x \log 3$

m1

$$x = \frac{\lg 4}{\lg 3} = 1.26185 \dots = 1.2619 \text{ to 4dp}$$

Accept 4dp or better

[If using T&I a full justification is required; else M0m0A0]

A1

4

(d) $f(x) = 3^{-x} + 1$

B1

1

[12]

Q30.

(a) $2e^x = 5$

$$e^x = \frac{5}{2}$$

(exact)

M1

$$x = \ln \frac{5}{2} \quad (0.916)$$

(A0 if further wrong work)

A1

2

(b) (i) $2e^x + 5e^{-x} = 7$
 $2e^{2x} + 5 = 7e^x$

Dealing with e^{-x}

M1

$$2y^2 - 7y + 5 = 0$$

AG

A1

2

(ii) $(2y - 5)(y - 1) = 0$

attempt to solve $y = \frac{5}{2}, 1$ (SC B1)

M1

$$x = \ln \frac{5}{2}$$

$$e^x = \frac{5}{2}$$

A1

$$x = 0 \text{ (or } \ln 1)$$

$$e^x = 1$$

A1

3

[7]